

UAV-UAV Communication Under Malicious Jamming: Trajectory Optimization with Turning Angle Constraint

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Abstract—In this paper, we investigate the UAV-UAV communication system, wherein a source UAV transmits signals to a destination UAV in the presence of a malicious jammer. We aim to minimize the flight time by optimizing the trajectory of UAVs with limited turning angle subject to the required minimum information to be transmitted. However, the formulated problem is non-convex and intractable to solve. Thus, we propose an iterative algorithm based on successive convex approximation (SCA) technique to solve it with the aid of Cauchy-Schwarz inequality and Young inequality. Numerical results show that the proposed algorithm can provide corresponding trajectory design with different jammer's deployment, and the freer turning angle along with the fewer required transmitted information can lead to shorter flight time.

Index Terms—UAV-UAV communication, anti-jamming, trajectory design, turning angle.

I. INTRODUCTION

Due to the high mobility, on-demand deployment and line-of-sight (LoS) channels, unmanned aerial vehicle (UAV)-enabled communication has attracted increasing attention recently [1]. The system performance can be significantly improved by providing a new degree of freedom for resource allocation via trajectory optimization.

Prior studies concerning trajectory optimization mainly focus on UAV-to-ground (U2G) communications, where UAVs are applied into uses as aerial base stations, relays or cellular-connected users to enhance the terrestrial communications [2–4]. However, as an essential part of multi-UAV systems, the UAV-UAV communication has not been fully investigated. In [5], the UAV-UAV communication was studied wherein the flight time was minimized via trajectory optimization in the presence of interference from terrestrial transmitters.

Meanwhile, the broadcasting nature of the radio propagation makes the UAV communication particularly vulnerable to jamming attacks [6]. Once the communication links are jammed, the connections will be degraded or even unavailable. Conventional anti-jamming studies mainly focus on the power domain, frequency domain and antenna spatial domain with transmit power optimization, frequency hopping, and receive beamforming. With UAV's highly controllable maneuverability, it is also appealing to defend against the wireless jamming attack in the spatial domain. In [7], the throughput of UAV was maximized by retreating away from the jammer with joint

trajectory and transmit power optimization. In [8], the signal-to-interference-plus-noise ratio (SINR) of destination node was optimized via trajectory design.

On the other hand, UAVs usually subject to turning angle constraint that they can not change flying direction arbitrarily especially for fixed wing UAVs. In fact, the turning angle is an important factor that impacts the performance of the UAV communication system deeply [9]. In [10], the turning angle constrained UAV-to-UAV communication was investigated, the trajectory and transmit power were optimized to minimize the flight time without taking the jamming signals into consideration. Note that although the UAV-UAV communication with interference was investigated in [5], the turning angle constraint was not considered. To the best of our knowledge, the trajectory optimization with turning angle constraint in UAV-UAV communication under malicious jamming is still an open problem.

Motivated by the above, in this paper, we study the flight time minimization problem of UAV-UAV communication subject to the required minimum information to be transmitted by optimizing the trajectory with the limited turning angle in the presence of jamming signals. Its main contributions are summarized as follows.

- We formulate a general optimization framework for the UAV-UAV communication in the presence of jamming signals, which considers the trajectory of the UAV with limited turning angle, the minimum information to be transmitted and aims to minimize the flight time.
- We propose an iterative algorithm to solve the formulated non-convex problem. Specifically, with the aid of slack variables, Cauchy-Schwarz inequality and Young inequality, the non-convex constraints are reformulated into convex constraints. By leveraging the successive convex approximation (SCA) technique, the reformulated problem is solved in an iterative manner until the convergence condition is reached.
- Numerical results show that the proposed algorithm can provide corresponding trajectory design to minimize the flight time with different jammer's deployment, tuning angle constraint and required minimum information to be transmitted. Moreover, a freer turning angle and a smaller amount of required information to be transmitted can lead

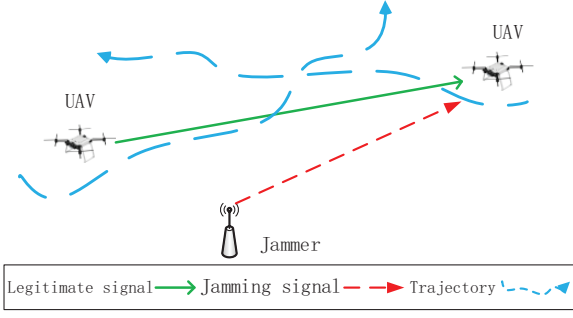


Fig. 1: UAV-UAV communication under malicious jamming.

to shorter flight time.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As shown in Fig. 1, we consider the UAV-UAV communication consisting of two UAVs, where the source UAV u_s transmit information to the destination UAV u_d while a jammer m is sending jamming signals to jam the transmission. Let $\mathcal{U} = \{u_s, u_d\}$ denotes the set of UAVs. We assume that both UAVs fly in a fixed altitude H , which is the minimum altitude to avoid obstacles. Without loss of generality, a three-dimensions (3D) Cartesian coordinate system is considered. Thus, the location of u_s , u_d and m are denoted as $\mathbf{q}_{u_s} = [x_{u_s}, y_{u_s}, H]^T \in \mathbb{R}^{3 \times 1}$, $\mathbf{q}_{u_d} = [x_{u_d}, y_{u_d}, H]^T \in \mathbb{R}^{3 \times 1}$ and $\mathbf{q}_m = [x_m, y_m, 0]^T \in \mathbb{R}^{3 \times 1}$, respectively. u_s , u_d are dispatched to fly from a given start point $\mathbf{q}_u^{\text{start}} = [x_u^{\text{start}}, y_u^{\text{start}}, H]^T \in \mathbb{R}^{3 \times 1}$, $u \in \mathcal{U}$ to an endpoint $\mathbf{q}_u^{\text{end}} = [x_u^{\text{end}}, y_u^{\text{end}}, H]^T \in \mathbb{R}^{3 \times 1}$, $u \in \mathcal{U}$ to execute the communication task.

To facilitate UAV trajectory optimization, path discretization [5, 10] method is adopted and then the trajectory of UAV is denoted by the consecutive waypoints $\mathcal{N} \in \{1, 2, \dots, n, \dots, N\}$. Thus, the trajectory of u_s and u_d can be approximately denoted by the combination of discrete locations, i.e., $\mathbf{q}_u[n] = [x_u[n], y_u[n], H]^T \in \mathbb{R}^{3 \times 1}$, $u \in \mathcal{U}$. Denote $t_L[n]$ as the time slot used by UAV from waypoint n to $n+1$. Thus, the total flight time can be denoted as $T = \sum_{n=1}^N t_L[n]$.

To guarantee an almost continuous trajectory for UAVs, the flying distance within each time slot is limited by $\Delta \max$ that

$$\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \leq \Delta \max, n = 2, \dots, N, u \in \mathcal{U}. \quad (1)$$

Considering the limited mobility of UAV, with $\phi_u[n]$, $\forall n, u \in \mathcal{U}$ denotes the turning angel of UAVs in each time slot, we have

$$\mathbf{q}_u[1] = \mathbf{q}_u^{\text{start}}, u \in \mathcal{U}, \quad (2)$$

$$\mathbf{q}_u[n] = \mathbf{q}_u^{\text{end}}, u \in \mathcal{U}, \quad (3)$$

$$\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \leq V_{\max} t_L[n], n=2, \dots, N, u \in \mathcal{U}, \quad (4)$$

$$V_{\min} t_L[n] \leq \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|, n=2, \dots, N, u \in \mathcal{U}, \quad (5)$$

$$\cos(\phi_u[n]) \leq \cos(\phi_{\max}), \forall n, u \in \mathcal{U}, \quad (6)$$

where $V_{\min}$, $V_{\max}$ and $\phi_{\max}$ denote the minimum flying speed, maximum flying speed and maximum turning angle of UAVs, respectively. Particularly, $\phi_{\max} = 180^\circ$ indicates that u is a rotate wing UAV that can change its flying direction freely.

Meanwhile, to avoid collision between UAVs, the minimum distance constraint is required and expressed as

$$\|\mathbf{q}_{u_s}[n] - \mathbf{q}_{u_d}[n]\| \geq d_{\min}, \forall n. \quad (7)$$

The field trial have verified that the U2G communication channels are mainly dominated by the large scale path-loss [11]. Hence, the channel power gains from u_s to u_d and m to u_d can be denoted as $g_{u_d, u_s}[n] = \beta_0 \|\mathbf{q}_{u_d}[n] - \mathbf{q}_{u_s}[n]\|^{-2}$, $g_{u_d, m}[n] = \beta_0 \|\mathbf{q}_{u_d}[n] - \mathbf{q}_m\|^{-2}$, respectively, where β_0 is the channel power gain at the reference distance $d_0 = 1$ m. Thus, the throughput from u_s to u_d in each waypoint is given by

$$R_{u_d, u_s}[n] = B t_L[n] \log_2 \left(1 + \frac{P_{u_s} g_{u_d, u_s}[n]}{P_m g_{u_d, m}[n] + \sigma^2} \right), \forall n, \quad (8)$$

where P_{u_s} , P_m and σ^2 denote the transmit power of u_s , transmit power of m and the power of additive white Gaussian noise (AWGN), respectively.

We aim to minimize the flight time by jointly optimizing the trajectory of u_s and u_d $\mathbf{q}_u[n]$, $u \in \mathcal{U}$ and time slots between each waypoints $t_L[n]$, with the minimum information requirement

$$\sum_{n=1}^N R_{u_d, u_s}[n] \geq Q_{\min}. \quad (9)$$

The problem can be mathematically formulated as

$$\begin{aligned} \text{(P1)} : \quad & \min_{\mathbf{q}_u[n], t_L[n]} \sum_{n=1}^N t_L[n] \\ \text{s.t.} \quad & (1) - (7), (9). \end{aligned} \quad (10)$$

Problem (P1) is difficult to solve due to the non-convexity of constraints (5)-(7) and (9). In the sequel, an efficient algorithm is proposed to solve it with the aid of slack variables, Cauchy-Schwarz inequality, Young inequality and the successive convex approximation (SCA) technique.

III. PROPOSED ALGORITHM

First, we use the following lemma to transform problem (P1) into a more trackable form.

Lemma 1: By introducing slack variables $G[n]$, $L[n]$ and $I[n]$, problem (P1) can be transformed into an equivalent optimization problem

$$\text{(P2)} : \quad \min_{\mathbf{q}_u[n], t_L[n]} \sum_{n=1}^N t_L[n] \quad (11)$$

$$\text{s.t.} \quad \sum_{n=1}^N G^2[n] \geq Q_{\min}, \quad (12)$$

$$G^2[n] \leq Bt_L[n] \log_2 \left(1 + \frac{1}{L[n]I[n]} \right), \forall n, \quad (13)$$

$$P_{u_s} g_{u_d, u_s}[n] \geq L[n]^{-1}, \forall n, \quad (14)$$

$$P_m g_{u_d, m}[n] + \sigma^2 \leq I[n], \forall n, \quad (15)$$

$$(1) - (7).$$

Proof: Denote the optimal value of objective function of problems (P1) and (P2) as p_1^* and p_2^* . On one hand, the introduction of $G[n]$, $L[n]$ and $I[n]$ enlarges the feasible region of problem (P2). Thus, problem (P2) provides a lower bound for problem (P1), i.e., $p_1^* \geq p_2^*$. On the other hand, assume that there exists certain points that satisfies $G^2[n] < Bt_L[n] \log_2 (1 + 1/(L[n]I[n]))$, $P_{u_s} g_{u_d, u_s}[n] > L[n]^{-1}$ and $P_m g_{u_d, m}[n] + \sigma^2 < I[n]$. We can always improve the value of $G[n]$ or decrease the value of $L[n]$ or $I[n]$ to makes them hold with equalities so that p_2^* can approach to p_1^* , i.e., $p_1^* \leq p_2^*$. Thus, **Lemma 1** is proved. ■

Note that constraints (12), (13), (15) are non-convex. Then, with the first-order Taylor expansion of $G^2[n]$, we transform (12) into convex as

$$\sum_{n=1}^N 2G_k[n]G[n] - G_k^2[n] \geq Q_{\min}. \quad (16)$$

Further, we use the following lemma to transform (13) into convex constraint.

Lemma 2: With any feasible point $(L^f[n], I^f[n])$, constraint (13) can be reformulated as

$$\frac{G^2[n]}{Bt_L[n]} \leq \log_2(1 + 1/L^f[n]I^f[n]) + A(L[n] - L^f[n]) + B(I[n] - I^f[n]), \quad (17)$$

where $A = -\log_2 e / (L^f[n] + (L^f[n])^2 I^f[n])$ and $B = -\log_2 e / (I^f[n] + (I^f[n])^2 L^f[n])$.

Proof: Since $f(x, y) = \log_2(1 + 1/xy)$ is a convex function with respect to x and y , its first-order Taylor expansion provides a global under-estimator at a given feasible point (x^f, y^f) , i.e.,

$$\begin{aligned} \log_2(1 + 1/xy) &\geq \log_2(1 + 1/x^f y^f) \\ &\quad - (x - x^f) \log_2 e / (x^f + (x^f)^2 y^f) \\ &\quad - (y - y^f) \log_2 e / (y^f + (y^f)^2 x^f). \end{aligned} \quad (18)$$

Thus, by applying $x = L[n]$ and $y = I[n]$, **Lemma 2** is proved. ■

Then, slack variable $d_m[n]$ is introduced to reformulate (15) as the following equivalent constraints

$$P_m \beta_0 d_m[n]^{-1} + \sigma^2 \leq I[n], \forall n, \quad (19)$$

$$d_m[n] \leq \|\mathbf{q}_{u_d}[n] - \mathbf{q}_m\|^2, \forall n, \quad (20)$$

$$d_m[n] \geq 0, \forall n. \quad (21)$$

The equivalence can be proved by contradiction that when the optimal solution is obtained, (20) should hold with equality, otherwise, we can always increase $d_m[n]$ to improve the objective function's value.

Note that (20) is non-convex, with the first-order Taylor expansion of $\|\mathbf{q}_{u_d}[n] - \mathbf{q}_m\|^2$, we derive the lower bound of $\|\mathbf{q}_{u_d}[n] - \mathbf{q}_m\|^2$ as

$$\begin{aligned} q_{u_d, m}^l[n] &= 2x_{u_d}^f[n]x_{u_d}[n] - (x_{u_d}^f[n])^2 + x_m^2 - 2x_m x_{u_d}[n] \\ &\quad + 2y_{u_d}^f[n]y_{u_d}[n] - (y_{u_d}^f[n])^2 + y_m^2 - 2y_m y_{u_d}[n], \end{aligned} \quad (22)$$

with which, (20) can be reformulated as a convex constraint

$$d_m[n] \leq q_{u_d, m}^l[n]. \quad (23)$$

Then, we use the following lemma to transform constraints (5) and (7) into convex

Lemma 3: Constraints (5) and (7) can be reformulated as

$$\frac{(\mathbf{q}_u[n] - \mathbf{q}_u[n-1])(\mathbf{q}_u[n] - \mathbf{q}_u[n-1])^T}{\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|} \geq t_L[n] V_{\min}, \forall n, u \in \mathcal{U}, \quad (24)$$

and

$$\frac{(\mathbf{q}_{u_d}[n] - \mathbf{q}_{u_s}[n])(\mathbf{q}_{u_d}[n] - \mathbf{q}_{u_s}[n])^T}{\|\mathbf{q}_{u_d}[n] - \mathbf{q}_{u_s}[n]\|} \geq d_{\min}, \forall n, \quad (25)$$

respectively.

Proof: With Cauchy-Schwarz inequality, for $\mathbf{a}$ and $\mathbf{b}$, we have $\|\mathbf{a}\| \|\mathbf{b}\| \geq |\langle \mathbf{a}, \mathbf{b} \rangle| \geq \langle \mathbf{a}, \mathbf{b} \rangle$. Then, let $\mathbf{b} = \mathbf{a}_f$, where $\mathbf{a}_f$ is a feasible point of $\mathbf{a}$, we have

$$\|\mathbf{a}\| \|\mathbf{a}_f\| \geq \langle \mathbf{a}, \mathbf{a}_f \rangle. \quad (26)$$

Thus, by applying $\mathbf{a} = \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|$, $u \in \mathcal{U}$ and $\mathbf{a} = \|\mathbf{q}_{u_d}[n] - \mathbf{q}_{u_s}[n]\|$, **Lemma 3** is proved. ■

Let $\mathbf{z}_u[n] = \mathbf{q}_u[n] - \mathbf{q}_u[n-1]$, $n = 2, \dots, N$, $u \in \mathcal{U}$, the maximum turning angel constraint (6) can be reformulated as $\frac{\mathbf{z}_u[n](\mathbf{z}_u[n-1])^T}{\|\mathbf{z}_u[n]\| \|\mathbf{z}_u[n-1]\|} \geq \cos \phi^{\max}$, $n = 2, \dots, N$, $u \in \mathcal{U}$, which also can be written as

$$\begin{aligned} \mathbf{z}_u[n](\mathbf{z}_u[n-1])^T \\ - \cos \phi^{\max} \|\mathbf{z}_u[n]\| \|\mathbf{z}_u[n-1]\| \geq 0, n = 2, \dots, N, u \in \mathcal{U}. \end{aligned} \quad (27)$$

Then, we use the following proposition [Proposition 3, 12] to handle the first term $\mathbf{z}_u[n](\mathbf{z}_u[n-1])^T$ in (27).

Proposition 1: For any given feasible point $\mathbf{a}_f$, $\mathbf{b}_f$, $\mathbf{a}\mathbf{b}^T$ is lower bounded by a concave function

$$\mathbf{a}\mathbf{b}^T \geq \mathbf{a}_f \mathbf{b}^T + \mathbf{a} \mathbf{b}_f^T - \mathbf{a}_f \mathbf{b}_f^T - \frac{1}{2} \|\mathbf{a} - \mathbf{a}_f\|^2 - \frac{1}{2} \|\mathbf{b} - \mathbf{b}_f\|^2 \quad (28)$$

Proof: Please refer to Appendix C in [12]. ■

Moreover, with Young inequality $ab \leq \frac{1}{2}(\varepsilon a^2 + \varepsilon^{-1} b^2)$ and $\varepsilon = \frac{\|\mathbf{z}_u^f[n-1]\|}{\|\mathbf{z}_u^f[n]\|}$ [10], we have

$$\begin{aligned} -\cos \phi^{\max} \|\mathbf{z}_u[n]\| \|\mathbf{z}_u[n-1]\| \geq \\ -\frac{1}{2} \cos \phi^{\max} \left(\varepsilon \|\mathbf{z}_u[n]\|^2 + \varepsilon^{-1} \|\mathbf{z}_u[n-1]\|^2 \right), u \in \mathcal{U}. \end{aligned} \quad (29)$$

Thus, with (28) and (29), the turning angel constraint (6) is reformulated as

$$\mathbf{a}_f \mathbf{b}^T + \mathbf{a} \mathbf{b}_f^T - \mathbf{a}_f \mathbf{b}_f^T - \frac{1}{2} \|\mathbf{a} - \mathbf{a}_f\|^2 - \frac{1}{2} \|\mathbf{b} - \mathbf{b}_f\|^2 - \frac{1}{2} \cos \phi^{\max} \left(\epsilon \|\mathbf{a}\|^2 + \epsilon^{-1} \|\mathbf{b}\|^2 \right) \geq 0, n=2, \dots, N, \quad (30)$$

where $\mathbf{a} = \mathbf{z}_u[n]$, $\mathbf{b} = \mathbf{z}_u[n-1]$, $\mathbf{a}_f = \mathbf{z}_u^f[n]$ and $\mathbf{b}_f = \mathbf{z}_u^f[n-1]$. Hence, problem (P2) is reformulated as

$$(P3) : \min_{\Phi[n]} \sum_{n=1}^N t_L[n] \quad (31)$$

s.t. (1) – (4), (14), (16), (17), (19), (21), (23) – (25), (30),

where $\Phi[n] = \{\mathbf{q}_u[n], t_L[n], G[n], L[n], I[n]\}$. Problem (P3) is a standard convex optimization problem, which thus can be optimally solved via CVX [13].

To sum up, with the introduced slack variables, Cauchy-Schwarz inequality and the Young inequality, the non-convex constraints are transformed into convex constraints. By leveraging the SCA technique, the overall algorithm solves the reformulated problem in an iterative way until the fractional decrease of the objective value is below a small threshold μ . The obtained solution of each iteration is non-increasing while the value of the problem is finite. Thus, the proposed algorithm is guaranteed to converge. The details of the proposed algorithm are presented in **Algorithm 1**.

Algorithm 1 The proposed algorithm for solving problem (P1)

- 1: **Initialization:** Denote the feasible solution in i_{th} iteration as $\Phi^i[n] = \{\mathbf{q}_u^i[n], t_L^i[n], G^i[n], L^i[n], I^i[n]\}$, set $i = 0$ as the initial feasible point.
 - 2: **Repeat**
 - 3: With given $\Phi^i[n]$, compute $\Phi^*[n]$ by solving problem (P3).
 - 4: $\Phi^{i+1}[n] = \Phi^*[n]$.
 - 5: Update $i \leftarrow i + 1$
 - 6: **Until** The fractional decrease of the objective value is below a small threshold μ .
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IV. NUMERICAL RESULTS

In this section, numerical results are provided to verify the performance of the proposed trajectory optimization algorithm. The parameters are set as follows. The communication bandwidth is $B = 1$ MHz. The noise power spectrum density is $N_0 = -169$ dBm/Hz. Thus, the corresponding noise power is $\sigma^2 = N_0 B = -109$ dBm. The transmit power of u_s and m are $P_{u_s} = 0.1$ W and $P_m = 0.1$ W, respectively. The altitude of level flight of UAV is $H = 100$ m. The minimum and maximum speed of UAV are $V_{\min} = 3$ m/s and $V_{\max} = 20$ m/s, respectively. The minimum distance between UAVs is $d_{\min} = 50$ m. The maximum flying distance within each time slot is $\Delta_{\max} = 30$ m. The channel power gain at the reference distance $d_0 = 1$ m is $\beta_0 = -60$ dB. The convergence threshold is $\mu = 10^{-3}$. Without loss of generality, we set $\mathbf{q}_{u_s}^{\text{start}} = (-100, 200, H)$ m, $\mathbf{q}_{u_s}^{\text{end}} = (100, 200, H)$ m, $\mathbf{q}_{u_d}^{\text{start}} = (-100, 0, H)$ m and $\mathbf{q}_{u_d}^{\text{end}} = (100, 0, H)$ m.

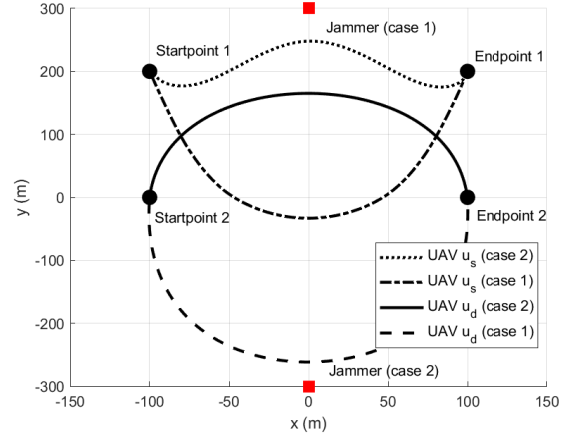


Fig. 2: Trajectories of UAVs with different jammer's location.

Fig. 2 shows the trajectories of UAVs when $\phi_{\max} = 180^\circ$ with different jammer's location that $\mathbf{q}_m = (0, 300, 0)$ m is set as case 1 and $\mathbf{q}_m = (0, -300, 0)$ m is set as case 2. It is observed that in both cases, UAV u_d flies away from the jammer as far as possible. This is because a further distance from the jammer can reduce the impact of jamming signals so as to improve the throughput. Meanwhile, UAV u_s tends to fly close to UAV u_d and keep in a stable distance. This is because a nearer distance between UAV u_d and UAV u_s can lead to better channel conditions so as to improve the throughput. However, UAVs subject to the minimum distance constraint to avoid collision so that they can only keep in a minimum distance instead of flying closer. Thus, the proposed algorithm is verified that it can adjust the trajectory with different jammer's deployment to combat jamming signals.

Fig. 3 shows the trajectories in case 1 with different required minimum information to be transmitted when $\phi_{\max} = 180^\circ$. It is observed that with the increase of required information, the trajectory of UAVs shifts to the opposite side of the jammer. This is because with a larger amount of required information to be transmitted, the UAV u_d has to fly further away from the jammer so as to reduce the impact of jamming signals, while the UAV u_s always tends to fly close to UAV u_d with the minimum distance between UAVs.

Fig. 4 shows the trajectories with different maximum turning angles in case 2. It is observed that with different $\phi_{\max}$, UAV u_d always flies away from the jammer and UAV u_s always flies close to UAV u_d so as to improve the throughput. However, the trajectory of UAVs becomes smoother with the decrease of $\phi_{\max}$. For instance, the trajectory of UAV u_s when $\phi_{\max} = 180^\circ$ presents more obviously "S" shape while UAV u_s flies in almost a straight line when $\phi_{\max} = 30^\circ$. This is because the limited turning angle restricts the freedom of trajectory optimization.

Fig. 5 shows the optimal flight time obtained by the proposed algorithm with different maximum tuning angles and required minimum information to be transmitted in case 1 and

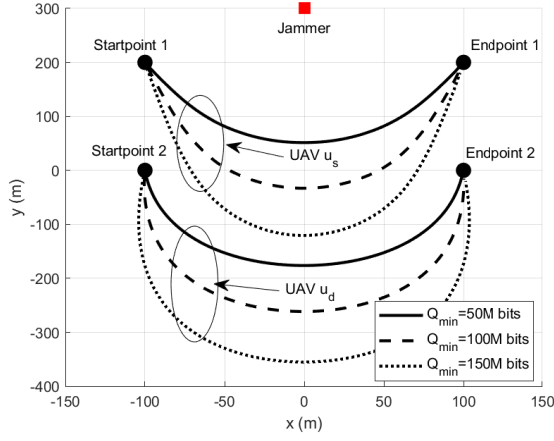


Fig. 3: Trajectories of UAVs with different required minimum information in case 1.

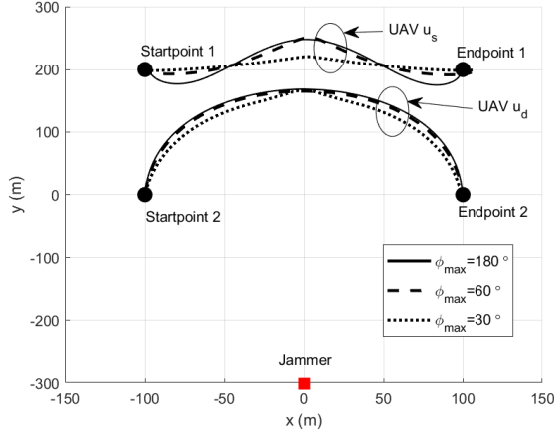


Fig. 4: Trajectories of UAVs with different maximum turning angles in case 2.

case 2. It is observed that with the increase of $Q_{\min}$, the flight time increases correspondingly for a longer time can enable more information exchange. Meanwhile, in both cases, the flight time increases with the decrease of the $\phi_{\max}$ due to the gradually restricted freedom of trajectory optimization. Hence, the proposed algorithm is further verified via performance comparison.

V. CONCLUSIONS

In this paper, we investigate the UAV-UAV communication under malicious jamming. By leveraging the SCA technique, an iterative algorithm is proposed to minimize the flight time via optimizing the trajectory of UAVs with turning angle constraint subject to the minimum information requirement. Numerical results show that the proposed algorithm can reduce the flight time significantly especially with freer turning angle and fewer required information exchange.

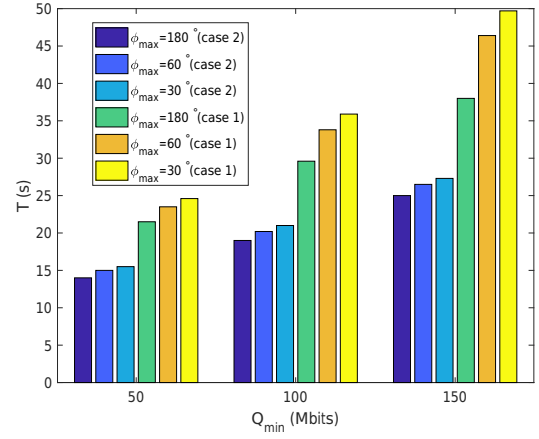


Fig. 5: Flight time with different required minimum information and maximum turning angle in both cases.

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